Bihar Engineering University, Patna

B.Tech 3rd Semester Examination, 2024

Course: B.Tech
Code: 100313

Subject: Object Oriented Programming using C++

Time: 03 Hours
Full Marks: 70

Instructions:-

(i) The marks are indicated in the right-hand margin.
*(ii) There are **NINE** questions in this paper.*
*(iii) Attempt **FIVE** questions in all.*
*(iv) Question No. **1** is compulsory.*

Q.1 ***Choose the correct option / answer the following (Any seven question only):*** **[2 x 7 = 14]**

(a) What is the default access specifier for class members in C++?

(i) public
(ii) protected
(iii) private
(iv) internal

(b) What will be the output of the following code?

```
#include <iostream>
class Test {
public:
  Test() { std::cout << "Constructor"; }
  ~Test() { std::cout << "Destructor"; }
};
int main() {
  Test t;
  return 0;
}
```

(i) Constructor
(ii) ConstructorDestructor
(iii) Destructor
(iv) DestructorConstructor

(c) What will be the output of the following code?

```
#include <iostream>
void display(int x) { std::cout << "Int"; }
void display(double x) { std::cout << "Double"; }
int main() {
  display(5.5f);
}
```

(i) Int
(ii) Float
(iii) Double
(iv) Error

(d) What is the output of the following code snippet?

```
int arr[] = {1, 2, 3};
int* p = arr;
cout << *(p + 1);
```

(i) 1
(ii) 3
(iii) 2
(iv) Error

(e) What is the output of the following code snippet?

```
string s = "C++";
s += " Rocks";
cout << s;
```

(i) C++
(ii) C++ Rocks
(iii) Rocks
(iv) Error

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(f) Which of the following statement is incorrect?

(i) A static member function cannot be declared as virtual function

(ii) Static and non-static functions with identical signatures cannot coexist in the same class in C++.

(iii) A static member function cannot access the "this pointer"

(iv) A. A non static member function can access only non static data members of a class.

(g) Identify the correct function template syntax.

(i) template function<T> void show(T t);

(ii) function template<T> show(T t);

(iii) template<class> show(T t);

(iv) template<typename T> void show(T t);

(h) What will be the output of this code?

```
#include <iostream>
class Demo {
public:
  void show() { std::cout << "Hello"; }
};
int main() {
  Demo d, *ptr = &d;
  ptr->show();
}
```

(i) Hello

(ii) Error

(iii) show

(iv) None

(i) What happens when a derived class object is assigned to a base class pointer?

```
class Base { };
class Derived : public Base { };
int main() {
  Derived d;
  Base* b = &d;
}
```

(i) Error

(ii) Not allowed

(iii) Requires template

(iv) Upcasting; allowed

(j) What is the output of this code?

```
#include <iostream>
class A {
public:
  ~A() { std::cout << "Destructor "; }
};
int main() {
  try {
    A a;
    throw 1;
  } catch (...) {
    std::cout << "Caught";
  }
}
```

(i) Caught Destructor

(ii) Destructor Caught

(iii) Caught

(iv) Error

Q.2 (a) Explain the features of Object-Oriented Programming (OOP). How does OOP differ from the procedural programming paradigm? Also, list the advantages of using OOP in C++. [7]

(b) Differentiate between while, do-while, and for loops in C++. Write a program to print the Fibonacci series using any loop. Hint: Fibonacci series: 0, 1, 1, 2, 3, 5, 8, 13, 21…. [7]

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Q.3 (a) Write a C++ program using function overloading to calculate the area of a square, rectangle, and circle using three overloaded area() functions. Demonstrate all three in main() and explain how function overloading works. [7]

(b) Write a C++ program using the string class to input two strings, concatenate them, compare them, and find the length of the resulting string. [7]

Q.4 (a) Differentiate between a struct and a class in C++. Explain with examples how data hiding and default access specifiers differ between the two. [7]

(b) Write a C++ program to define a class Account with data members like accountNumber, holderName, and balance. Include a default constructor to initialize default values, a parameterized constructor to assign specific values, and a destructor to display a message when the object is destroyed. In the main() function, create objects using both constructors. Also, briefly explain the role of constructors and destructors in C++. [7]

Q.5 (a) Write a program to overload the + operator to add two complex numbers using a class Complex. Also, discuss the rules and limitations of operator overloading. [7]

(b) What is a function template in C++? Write a template function maximum() that returns the larger of two values. Demonstrate its use with integers, floats, and characters. [7]

Q.6 (a) What is a virtual base class in C++? Explain its role in resolving the diamond problem with an example involving classes A, B, C, and D, where D inherits from both B and C, and B and C inherit from A. [7]

(b) Explain single and multiple inheritance in C++ with suitable examples. Write a program to demonstrate both types of inheritance. Also, discuss how ambiguity is handled in multiple inheritance. [7]

Q.7 (a) Write a C++ program to demonstrate the use of an abstract class with a pure virtual function. Explain how this supports runtime polymorphism. [7]

(b) What is a void pointer in C++? Explain its use with an example. Also, demonstrate how a void pointer can be typecast to access different data types. [7]

Q.8 (a) Write a C++ program to demonstrate a pointer to a class and a pointer to an object. Create a class Student and use a pointer to access its member functions. Explain the concept of dynamic object access using pointers. [7]

(b) Explain how arrays and pointers are related in C++. Write a program to process an array using pointers (e.g., calculate the sum of elements). Discuss pointer arithmetic with arrays. [7]

Q.9 (a) What is exception handling in C++? Explain the purpose and working of the try, throw, and catch keywords with proper syntax. Also, briefly explain the concept of stack unwinding in the context of exceptions. [7]

(b) Write a program to handle division by zero using exception handling. Also, describe how control is transferred during an exception. [7]

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